· CLEAR CONCEPTS · VISUALS · PYTHON

Probability & Bayesian Networks

From uncertainty to intelligent reasoning

Learning outcomes

CORE IDEA

Understand how AI handles incomplete information.

1. Calculate simple and conditional probabilities.
2. Read and build Bayesian Networks.
3. Perform inference and implement examples in Python.

WORKED EXAMPLE

By the end: Given an alarm, estimate the probability of a real fault.

Why probability matters

CORE IDEA

AI rarely observes the world perfectly.

Sensors can be noisy.

Events have multiple possible causes.

The same evidence may support several explanations.

WORKED EXAMPLE

A smoke alarm sounds.

Was there a fire?

Was it cooking smoke?

Was it a false alarm?

Uncertainty versus certainty

CORE IDEA

A probability expresses how strongly a model supports an event.

P = 0: impossible under the model.

P = 1: certain under the model.

P = 0.7: 70% probability, not a guarantee.

WORKED EXAMPLE

P(rain tomorrow) = 0.7

Rain is likely, but sunshine is still possible.

Sample space and events

The sample space Ω contains all possible outcomes.

$$\Omega = \{1, 2, 3, 4, 5, 6\}$$

A = {2, 4, 6} = rolling an even number.

TAKEAWAY **An event is a subset of possible outcomes.**

Basic probability rules

Probabilities cannot be negative or greater than 1.

$$0 \le P(A) \le 1 \quad \text{and} \quad \Sigma\, P(\text{outcome}) = 1$$

Fair die: six outcomes, each with probability 1/6.

TAKEAWAY All mutually exclusive exhaustive outcomes sum to 1.

WORKED EXAMPLE · TASK CONDITION

Task 1 — Probability of a die roll

A fair six-sided die is rolled once.

GIVEN

Six equally likely outcomes:
1, 2, 3, 4, 5, 6

FIND

What is the probability of an even result?

SOLUTION PLAN

Identify the sample space.
Count favorable outcomes.
Divide 3 by 6.

Calculating event probability

For equally likely outcomes, count favorable cases.

P(A) = favorable outcomes / all outcomes

Even number on a fair die: 3 / 6 = 0.50.

TAKEAWAY **Probability of an even roll is 50%.**

Random variables

CORE IDEA

A random variable represents an uncertain quantity.

We use uppercase symbols such as T or F.

A variable has a set of possible values.

A distribution assigns a probability to each value.

WORKED EXAMPLE

T = temperature state

Values: {normal, high}

P(high) = 0.30

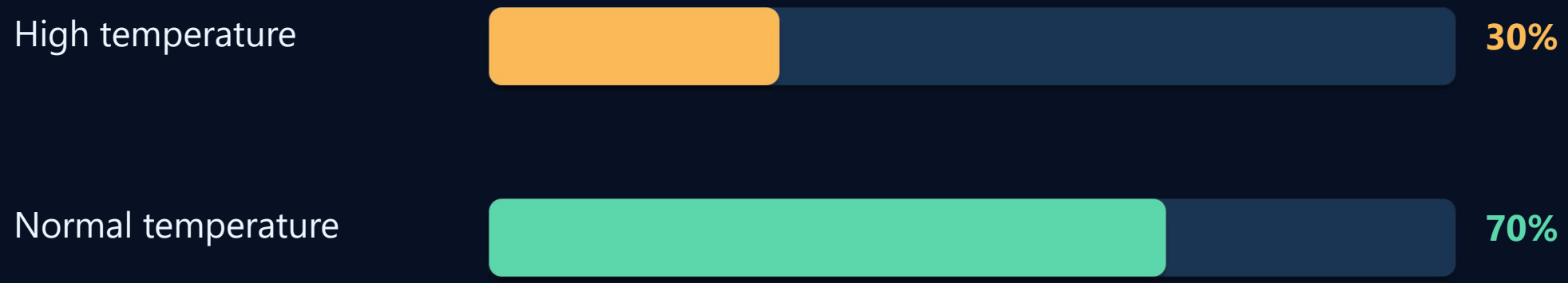
EXAMPLES

Probability distribution

Temperature state	Probability
Normal	0.70
High	0.30
TOTAL	1.00

Check: all probabilities add to 1.

A probability is a proportion



0.30 + 0.70 = 1.00 → all possible states are accounted for.

Read 0.30 as 30 out of every 100 comparable cases.

One variable versus two

CORE IDEA

A joint distribution describes combinations of variables.

P(T) describes temperature alone.

P(F) describes failure alone.

P(T,F) describes temperature AND failure.

WORKED EXAMPLE

T = high, F = yes

P(T=high,F=yes) = 0.18

WORKED EXAMPLE · TASK CONDITION

Task 2 — Machine failure and temperature

An industrial monitoring system records temperature and machine condition.

GIVEN

$P(\text{High},\text{Failure})=0.18$

$P(\text{High},\text{Healthy})=0.12$

$P(\text{Normal},\text{Failure})=0.02$

$P(\text{Normal},\text{Healthy})=0.68$

FIND

Find $P(\text{Failure})$ and $P(\text{Failure} \mid \text{High})$.

SOLUTION PLAN

First sum the failure cells.

Then divide the High-and-Failure cell by P(High).

EXAMPLES

Joint probability: equipment example

Temperature	Failure yes	Failure no
High	0.18	0.12
Normal	0.02	0.68

All four joint probabilities sum to 1.

Joint probability: see all four combinations

High + failure

18% of all cases

High + healthy

12% of all cases

Normal + failure

2% of all cases

Normal + healthy

68% of all cases

Each cell means the probability of BOTH the row and column conditions.

Reading a joint probability

An AND event means both conditions occur.

P(High AND Failure) = 0.18

Look at the cell in the High row and Failure=yes column.

TAKEAWAY **Joint probability concerns the intersection of events.**

Marginalization: sum out a variable

Add across values of variables we do not need.

$$P(F=yes) = \Sigma_t P(F=yes, T=t)$$

P(F=yes) = 0.18 + 0.02 = 0.20

TAKEAWAY **The overall probability of failure is 20%.**

Marginalization: add the matching cells

High + failure

18%

+

Normal + failure

2%

=

All failures

18% + 2% = 20%

To find overall failure probability, ignore temperature by adding both failure cells.

Another marginal probability

The same method works for temperature.

$$P(T=\text{high}) = \Sigma_f \; P(T=\text{high}, F=f)$$

P(T=high) = 0.18 + 0.12 = 0.30

TAKEAWAY The overall probability of high temperature is 30%.

Conditional probability

CORE IDEA

Evidence changes the population we are considering.

P(F | High) means probability of failure GIVEN high temperature.

We only look at cases where temperature is high.

WORKED EXAMPLE

Of the 30% high-temperature cases, 18% are also failures.

So 0.18 / 0.30 = 0.60.

Conditional probability formula

Divide joint probability by the probability of the condition.

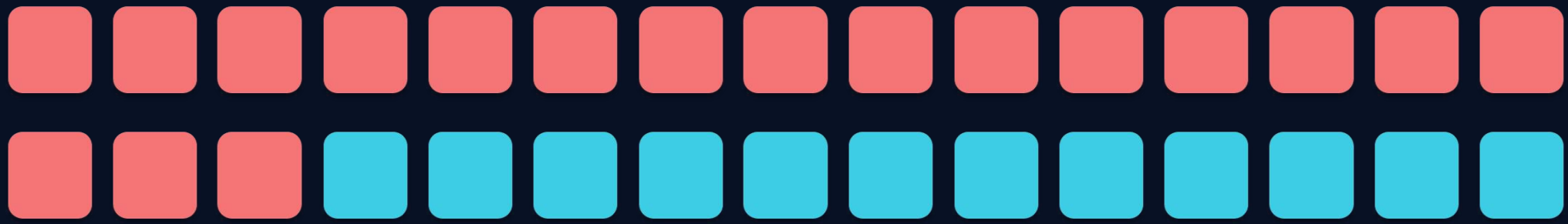
$$P(A \mid B) = P(A,B) / P(B)$$

P(Failure | High) = 0.18 / 0.30 = 0.60

TAKEAWAY **Given high temperature, failure probability is 60%.**

VISUAL EXPLANATION · CS188

Conditional probability: zoom into the high-temperature group



18 failures / 30 high-temperature cases = 0.60 = 60%

Red = failure · Blue = no failure

Among the 30 high-temperature cases per 100, 18 are failures.

Prior versus posterior

CORE IDEA

A prior is a belief before new evidence; a posterior is after.

Prior: P(Failure) = 20%.

Evidence: temperature is high.

Posterior: P(Failure | High) = 60%.

WORKED EXAMPLE

The same machine is now considered more likely to fail because we observed a warning sign.

Product rule

Rearrange conditional probability to obtain a joint probability.

$$P(A,B) = P(A \mid B) \times P(B)$$

P(Failure,High) = 0.60 × 0.30 = 0.18

TAKEAWAY **Multiply conditional probability by the condition's probability.**

Chain rule

Extend the product rule to several variables.

$$P(X,Y,Z) = P(X)P(Y|X)P(Z|X,Y)$$

Three-variable probabilities can be broken into smaller factors.

TAKEAWAY **This identity works for any joint distribution.**

Bayes' theorem: why it is useful

CORE IDEA

We often know the probability of evidence given a cause, but want the reverse.

Known: P(Alarm | Fault).

Wanted: P(Fault | Alarm).

Bayes' theorem reverses the conditioning.

WORKED EXAMPLE

A sensitive alarm does NOT mean that most alarms indicate real faults.

Bayes' theorem

Update a prior belief using evidence.

$$P(F|A) = P(A|F)P(F) / P(A)$$

Prior × likelihood ÷ evidence = posterior.

TAKEAWAY **This is the mathematical rule for belief updating.**

Bayes' Theorem

Update our belief about a cause after observing evidence.

Formula

$$P(A \mid B) = \frac{P(B \mid A)\,P(A)}{P(B)}$$

- $P(A \mid B)$: Probability of cause A given evidence B
- $P(B \mid A)$: Likelihood of evidence B if A is true
- $P(A)$: Prior probability of A
- $P(B)$: Overall probability of evidence B

Example: Machine Fault Detection

Problem statement

- A factory has many machines.
- A sensor can detect a fault, but it is not perfect.
- If the sensor shows an alarm, what is the probability that the machine is actually faulty?

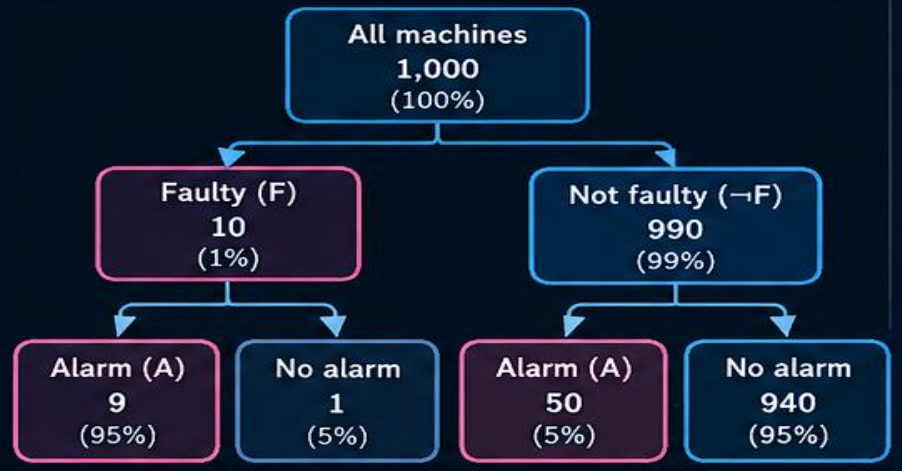
Given data

- $P(F) = 0.01$ (1% of machines are faulty)
- $P(A \mid F) = 0.95$ (sensor detects alarm when faulty)
- $P(A \mid \neg F) = 0.05$ (false alarm rate)

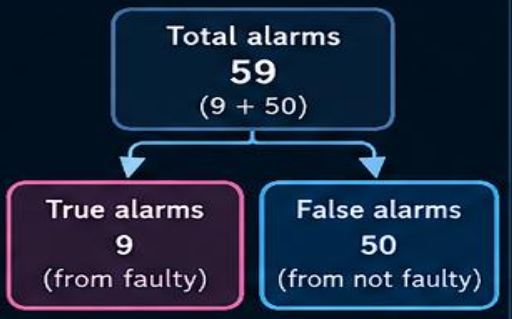
Task

- Calculate $P(F \mid A)$: the probability that the machine is faulty given that the sensor shows an alarm.

All machines (1,000 machines)



Observed alarm (A)



$$P(F \mid A) = \frac{9}{59} \approx 0.153 \approx 15.3\%$$

Step-by-step solution

1 Compute P(A)

$$\begin{aligned} P(A) &= P(A \mid F)P(F) + P(A \mid \neg F)P(\neg F) \\ &= 0.95 \times 0.01 + 0.05 \times 0.99 \\ &= 0.0095 + 0.0495 \\ &= 0.059 \end{aligned}$$

2 Apply Bayes' theorem

$$\begin{aligned} P(F \mid A) &= \frac{P(A \mid F)P(F)}{P(A)} \\ &= \frac{0.95 \times 0.01}{0.059} \\ &= \frac{0.0095}{0.059} \approx 0.153 \end{aligned}$$

3 Result

$$P(F \mid A) \approx 0.153 \approx 15.3\%$$

4 Interpretation

Even though the sensor is 95% accurate, an alarm does not mean the machine is certainly faulty. The probability of an actual fault is only 15.3% because false alarms are relatively common (5% of non-faulty machines).

WORKED EXAMPLE · PROBLEM STATEMENT

Task A — Diagnose a machine from an alarm

A factory monitors 1,000 machines with a binary fault alarm.

GIVEN

Fault rate = 5%

Sensitivity = 90%

False-alarm rate = 10%

FIND

If the alarm is ON, what is the probability of an actual fault?

PLAN

Count true and false alarms; then use Bayes' theorem.

Fault detector: given information

Quantity	Meaning	Value
P(F)	Prior fault probability	0.05
P(A|F)	Alarm when faulty	0.90
P(A|not F)	False-alarm rate	0.10

Question: What is P(F|A)?

Bayes example — step 1

First calculate the overall probability of an alarm.

$$P(A) = P(A|F)P(F) + P(A|\neg F)P(\neg F)$$

$= 0.90 \times 0.05 + 0.10 \times 0.95 = 0.14$

TAKEAWAY **An alarm occurs in 14% of cases.**

Bayes example — step 2

Now update the probability of a fault.

$$P(F|A) = 0.90 \times 0.05 / 0.14$$

$= 0.045 / 0.14 \approx 0.3214$

TAKEAWAY A positive alarm implies a 32.14% fault probability.

Why is it only 32.14%?

CORE IDEA

The base rate of faults is low.

Imagine 1,000 machines:

50 faulty → 45 positive alarms.

950 healthy → 95 false alarms.

WORKED EXAMPLE

Among 140 alarms, only 45 indicate actual faults.

45 / 140 = 32.14%.

Bayes' theorem with 1,000 machines

1,000 machines

50 faulty

950 healthy

Positive alarms

45 from faulty

95 from healthy

Among 140 alarms

45 true faults

95 false alarms

45 / 140 = 32.14%

Counts make the difference between sensitivity and posterior probability easy to see.

Python 1 — Bayes' theorem

PYTHON · RUN IN COLAB

```
p_fault = 0.05
p_alarm_fault = 0.90
p_alarm_healthy = 0.10

p_alarm = (p_alarm_fault*p_fault
          + p_alarm_healthy*(1-p_fault))
posterior = p_alarm_fault*p_fault/p_alarm
print(f'{posterior:.2%}')
```

LINE-BY-LINE LOGIC

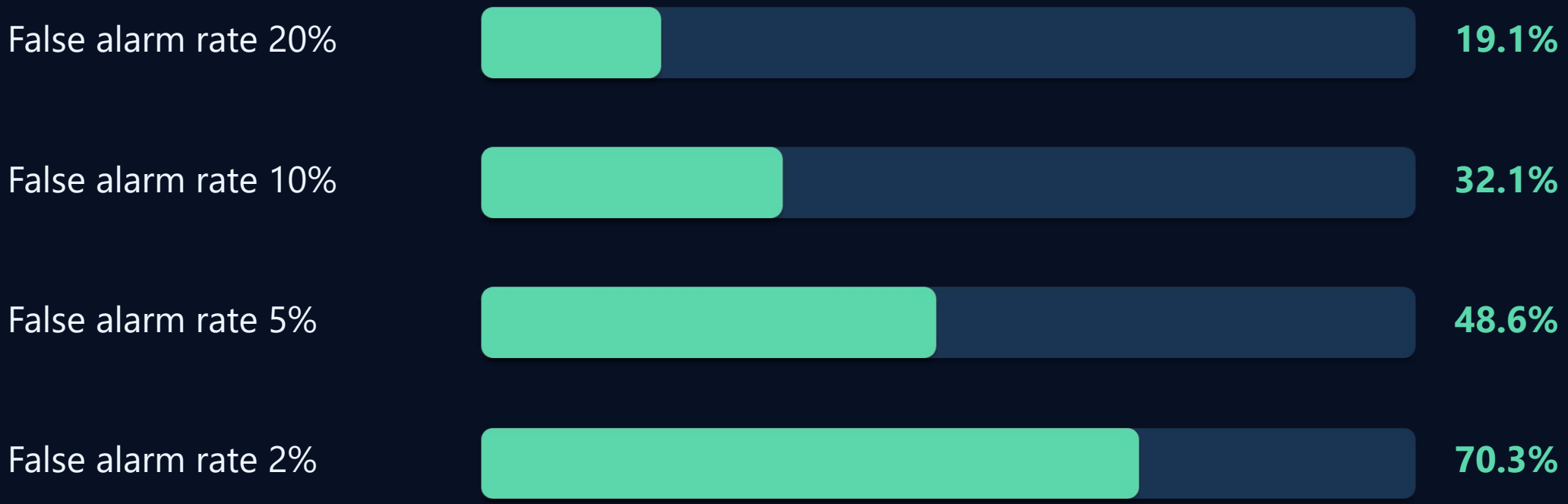
1. Define the prior.

2. Calculate the total alarm probability.

3. Divide the fault-and-alarm probability by all alarms.

Output: 32.14%

A better detector changes the posterior



With the same 5% fault prevalence and 90% sensitivity, reducing false alarms helps.

Independence

CORE IDEA

Two variables are independent when knowing one does not change the other.

For independent X and Y:

$P(X,Y) = P(X)P(Y)$.

Independent events can be modeled separately.

WORKED EXAMPLE

Two independent fair coin tosses:

$P(\text{heads,heads}) = 0.5 \times 0.5 = 0.25$.

Conditional independence

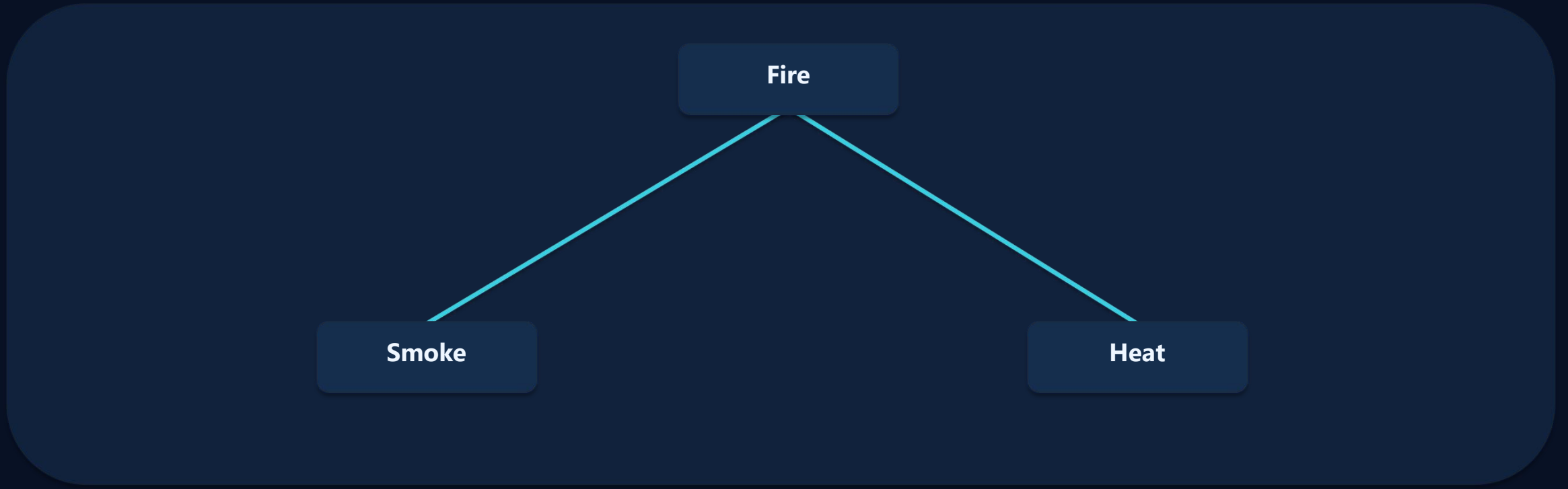
Two variables can become independent once a third is known.

$$P(X,Y \mid Z) = P(X \mid Z)P(Y \mid Z)$$

Smoke and heat may be conditionally independent given fire in a simplified model.

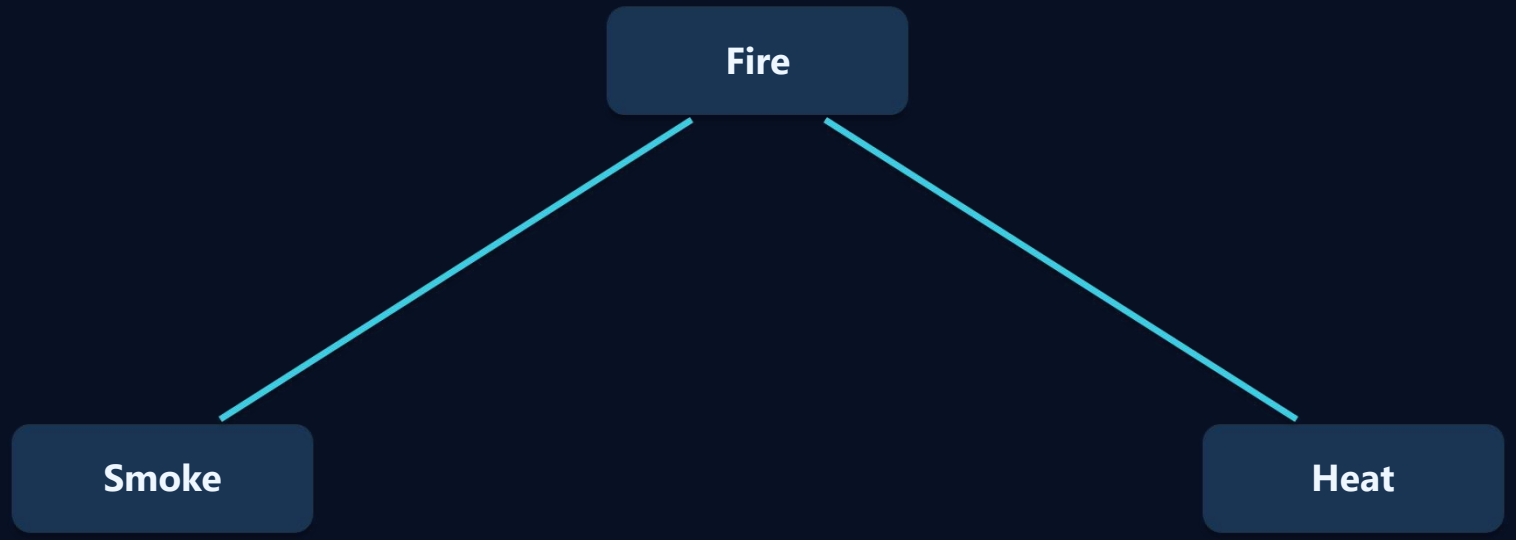
TAKEAWAY **Knowing the common cause can explain their correlation.**

Common-cause structure



Smoke and heat share a parent: Fire.

Conditional independence: a shared cause



$$P(\text{Smoke}, \text{Heat} \mid \text{Fire}) = P(\text{Smoke} \mid \text{Fire}) \times P(\text{Heat} \mid \text{Fire})$$

Smoke and heat may be correlated, but independent after the fire state is known (model assumption).

From probability to Bayesian Networks

CORE IDEA

A Bayesian Network encodes a joint distribution using a graph and local probabilities.

Node = random variable.

Arrow = modeled direct dependence.

CPT = probabilities conditioned on parent variables.

WORKED EXAMPLE

The graph structure helps avoid storing a giant joint probability table.

From Probability to Bayesian Networks

Building from simple probability concepts to powerful graphical models for reasoning under uncertainty.

1 Basic Probability

Quantify uncertainty

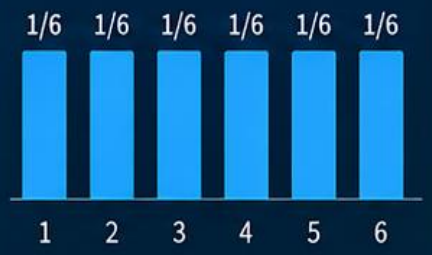
Definition

Probability assigns a number between 0 and 1 to an event.

$$0 \le P(A) \le 1$$

$$\sum_{\omega \in \Omega} P(\omega) = 1$$

Example: rolling a fair die



$$P(\text{roll is even}) = \frac{3}{6} = 0.5$$

2 Random Variables

Represent uncertain quantities

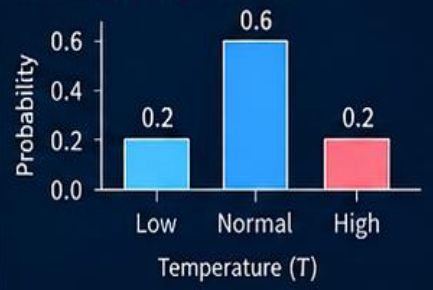
Definition

A random variable is a function that assigns a value to each possible world/outcome.

Example: Weather

Weather (W)	P(W)
Sunny	0.6
Rainy	0.2
Cloudy	0.2

Example: Temperature



3 Joint and Conditional Probability

Reason about multiple variables

Joint probability

Probability of two events together.

$$P(A, B) = P(A \cap B)$$

Example: Weather and Traffic

P(T,W)	Sunny	Rainy	Cloudy
Traffic (T)			
Yes	0.1	0.3	0.2
No	0.5	0.1	0.3

Marginalization

Sum out unwanted variables.

$$P(T) = \sum_{w} P(T, w)$$

Conditional probability

Probability of A given B.

$$P(A \mid B) = \frac{P(A, B)}{P(B)}$$

4 Bayes' Theorem

Update beliefs with new evidence

Formula

$$P(A \mid B) = \frac{P(B \mid A) P(A)}{P(B)}$$

Example: Medical Test

Problem

- 1% of people have the disease.
- Test detects 99% of diseased people.
- False positive rate is 5%.
- If the test is positive, what is the probability the person has the disease?

Calculation

$$P(D \mid +) = \frac{0.99 \times 0.01}{0.99 \times 0.01 + 0.05 \times 0.99}$$

$$= \frac{0.0099}{0.0594} \approx 0.167$$

$$\approx \mathbf{16.7\%}$$

Interpretation

Even with a highly accurate test, a positive result means only a **16.7%** chance of actually having the disease because the disease is rare.

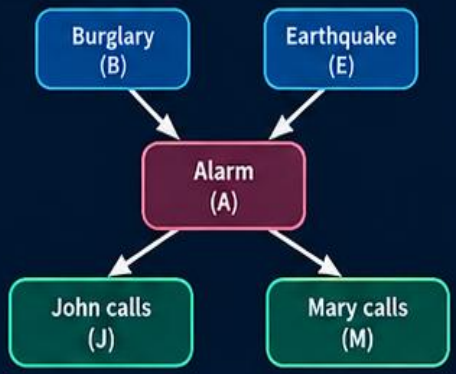
5 Bayesian Networks

Represent complex joint distributions efficiently

Key idea

Use a directed graph and local conditional probabilities to model the relationships between many variables.

Example: Alarm network



Joint distribution factorization

$$P(B, E, A, J, M) = P(B)P(E)P(A \mid B, E)P(J \mid A)P(M \mid A)$$

Conditional Probability Tables (CPTs)

$P(A \mid B, E)$	$E = T$	$E = F$
$B = T$	0.95	0.94
$B = F$	0.29	0.001

1. Represent uncertainty
Probability theory gives the basic language.

2. Combine information
Use conditional probability and Bayes' theorem to update beliefs.

3. Model complex domains
Bayesian Networks capture dependencies and enable efficient probabilistic inference.

WORKED EXAMPLE · PROBLEM STATEMENT

Task B — Infer a fire from an alarm

A fire can cause smoke, and smoke can activate an alarm.

GIVEN

P(F)=.02

P(S|F)=.90; P(S|¬F)=.05

P(A|S)=.95; P(A|¬S)=.01

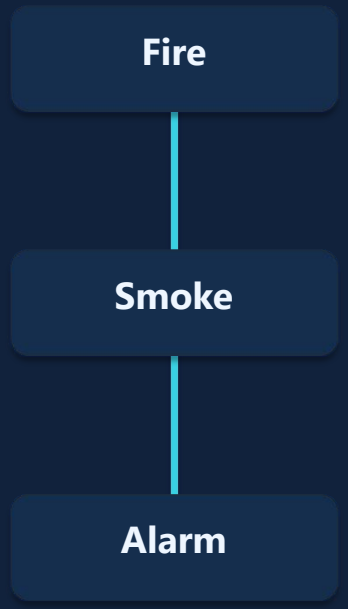
FIND

Calculate P(Fire | Alarm), even though Smoke is unknown.

PLAN

Sum over Smoke=yes/no; then normalize.

Simple Bayesian Network: fire alarm



The model assumes fire influences smoke, and smoke influences the alarm.

Parents and children

CORE IDEA

An arrow X → Y makes X a parent of Y.

Fire is parent of Smoke.

Smoke is parent of Alarm.

Alarm is a descendant of Fire.

WORKED EXAMPLE

A Bayesian Network must be directed and acyclic (no directed loops).

CPT 1 — Prior probability of fire

Fire	Probability
Yes	0.02
No	0.98

A root node has no parents; its CPT is a prior distribution.

CPT 2 — Smoke given fire

Fire	Smoke yes	Smoke no
Yes	0.90	0.10
No	0.05	0.95

Each row is a probability distribution conditioned on Fire.

CPT 3 — Alarm given smoke

Smoke	Alarm yes	Alarm no
Yes	0.95	0.05
No	0.01	0.99

The alarm can activate without smoke: a 1% false-alarm chance.

A CPT is a lookup table for a child variable

PARENT STATE	SMOKE = YES	SMOKE = NO
Fire = YES	90%	10%
Fire = NO	5%	95%

Fix the parent value, then read the probability of the child.

Bayesian Network factorization

Multiply each node's probability conditioned on its parents.

$$P(F,S,A) = P(F)P(S|F)P(A|S)$$

This is the network's compact representation of the joint distribution.

TAKEAWAY **One local factor per node.**

WORKED EXAMPLE · TASK CONDITION

Task 3 — Compute a joint probability

A fire can cause smoke, and smoke can activate an alarm.

GIVEN

P(Fire)=0.02

P(Smoke | Fire)=0.90

P(Alarm | Smoke)=0.95

FIND

Find P(Fire, Smoke, Alarm).

SOLUTION PLAN

Use the Bayesian Network factorization.

Multiply the three given probabilities.

Joint probability: complete example

Compute the probability of fire, smoke, and alarm together.

$$P(F=\text{yes}, S=\text{yes}, A=\text{yes})$$

$$= 0.02 \times 0.90 \times 0.95 = 0.0171$$

TAKEAWAY All three events occur together with 1.71% probability.

Inference: ask questions using evidence

CORE IDEA

Inference computes a desired probability from a probabilistic model.

Query: Is there a fire?

Evidence: Alarm=yes.

Hidden variable: Smoke (unknown).

WORKED EXAMPLE

We want P(Fire=yes | Alarm=yes).

Inference — sum out smoke

Alarm may activate whether smoke is present or absent.

$$P(A|F) = \Sigma_s \, P(A|s)P(s|F)$$

= 0.95×0.90 + 0.01×0.10 = 0.856

TAKEAWAY **The alarm sounds in 85.6% of fire cases.**

Inference: observe the alarm and reason backward



Observed: Alarm = YES

Question: How likely is Fire = YES?

The graph points from causes to effects; inference can work from effects back to causes.

Inference — consider no fire

Also calculate alarm probability if there is no fire.

$$P(A|\neg F) = \Sigma_s \, P(A|s)P(s|\neg F)$$

$= 0.95 \times 0.05 + 0.01 \times 0.95 = 0.057$

TAKEAWAY The alarm sounds in 5.7% of no-fire cases.

Inference — normalize

Combine fire and no-fire cases to calculate P(Alarm).

$$P(A)=0.856\times0.02 + 0.057\times0.98$$

P(A) = 0.07298

TAKEAWAY **Overall probability of an alarm is 7.298%.**

Inference — final posterior

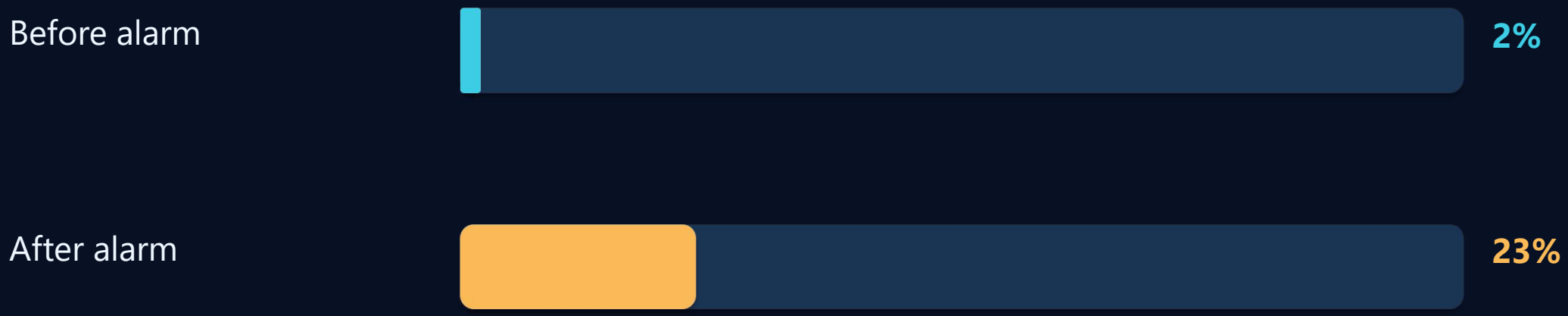
Apply Bayes' theorem using the computed likelihood.

$$P(F|A) = 0.856 \times 0.02 \,/\, 0.07298$$

≈ 0.2346

TAKEAWAY The fire probability rises from 2% to 23.46%.

Posterior probability: the update is visible



2.00% → 23.46% **(about 11.7 times the prior probability)**

The alarm increases the fire probability, but does not prove that a fire occurred.

Python 2 — inference by enumeration

PYTHON · RUN IN COLAB

```
p_f = 0.02
p_a_f = .90*.95 + .10*.01
p_a_nf = .05*.95 + .95*.01
p_a = p_a_f*p_f + p_a_nf*(1-p_f)
p_f_a = p_a_f*p_f/p_a
print(f'{p_f_a:.2%}')
```

LINE-BY-LINE LOGIC

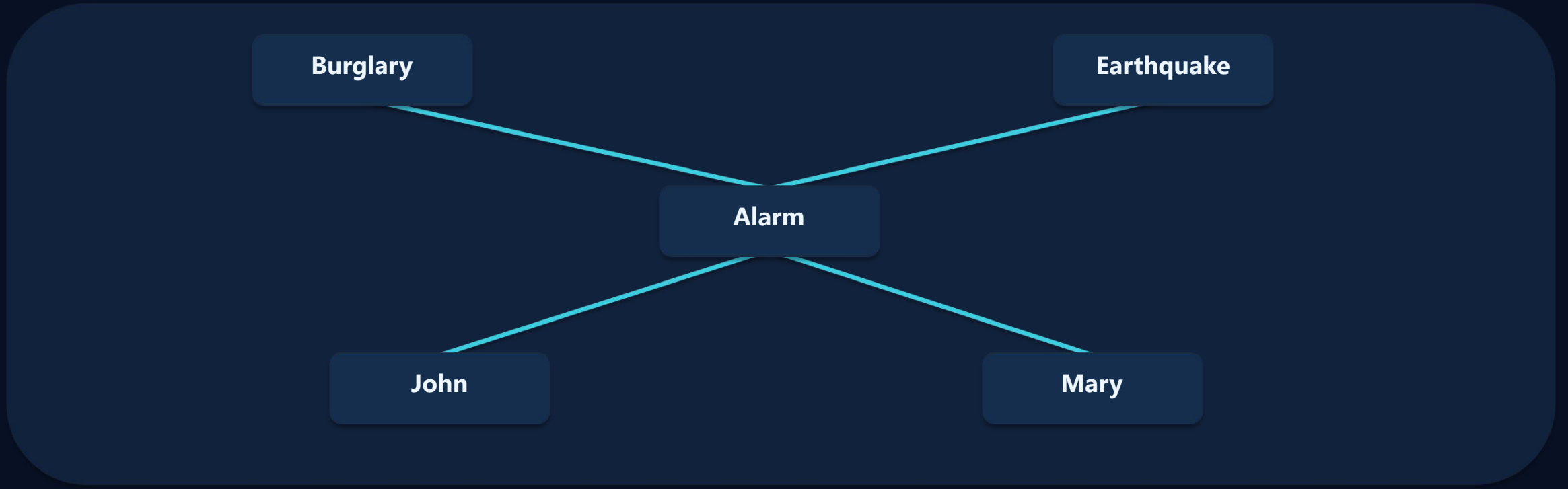
1. Sum over smoke states.

2. Calculate alarm probabilities with and without fire.

3. Normalize with Bayes.

Output: 23.46%

Classic burglary–alarm network



John and Mary are conditionally independent given the alarm state.

Burglary network factorization

Each variable is conditioned only on its parents.

$$P(B,E,A,J,M) = P(B)P(E)P(A|B,E)P(J|A)P(M|A)$$

This is the classic example from Lecture 14.

TAKEAWAY **A large joint model is represented using local CPTs.**

Explaining away

CORE IDEA

Two initially independent causes can become dependent after observing a common effect.

Alarm sounded: burglary becomes more plausible.

Then earthquake is confirmed: burglary may become less plausible.

WORKED EXAMPLE

Knowing an alternative explanation changes our beliefs.

Why Bayesian Networks scale better

CORE IDEA

A sparse network stores small conditional tables instead of every joint assignment.

Full joint distribution: $O(d^n)$.

Sparse BN with $\leq k$ parents: $O(n \cdot d^k)$.

Savings can be dramatic when k is small.

WORKED EXAMPLE

20 binary variables have 2^{20} = 1,048,576 joint assignments.

EXACT INFERENCE

What exactly is inference?

WHAT IT MEANS

Inference answers questions about unobserved variables using observed evidence.

QUERY = variable we want.

EVIDENCE = variable we observe.

HIDDEN = variables we must account for.

MINI EXAMPLE

Query: Fire?

Evidence: Alarm=ON

Hidden: Smoke

Answer: P(Fire | Alarm)

Exact inference: two major methods

WHAT IT MEANS

Exact inference returns the mathematically correct probability for the given model.

1. Enumeration: list and sum possible hidden states.
2. Variable elimination: combine small factors and reuse results.

MINI EXAMPLE

Fire → Smoke → Alarm

Compute P(Fire | Alarm=ON) using CPTs, not simulated data.

WORKED EXAMPLE · PROBLEM STATEMENT

Task C — Exact inference by enumeration

The fire–smoke–alarm network is given by the three CPTs.

GIVEN

P(F)=.02

P(S|F)=.90; P(S|¬F)=.05

P(A|S)=.95; P(A|¬S)=.01

FIND

Find P(F=YES | A=YES).

PLAN

Compute the joint probabilities for Smoke=yes and Smoke=no.

EXACT INFERENCE

Enumeration: list the hidden states

Smoke is hidden, so we consider both values.

$$P(F,A) = P(F,S,A) + P(F,\neg S,A)$$

$0.02 \times 0.90 \times 0.95 + 0.02 \times 0.10 \times 0.01$

$= 0.01710 + 0.00002 = 0.01712$

Numerator: $P(F,A) = 0.01712$

Enumeration: calculate the denominator

We also need all ways the alarm can sound.

$$P(A) = P(F,A) + P(\neg F,A)$$

$P(\neg F,A) = 0.98\times(0.05\times0.95 + 0.95\times0.01)$
$= 0.05586$; hence $P(A) = 0.07298$

Denominator: $P(A) = 0.07298$

EXACT INFERENCE

Enumeration: normalize the answer

Divide the fire-and-alarm probability by all alarm cases.

$$P(F \mid A) = P(F,A) / P(A)$$

$P(F \mid A) = 0.01712 / 0.07298 \approx 0.2346$

Final answer: approximately 23.46%

Python — exact enumeration

RUN IN PYTHON OR GOOGLE COLAB

```
from itertools import product
p_f = {True:.02, False:.98}
p_s = {True:{True:.90,False:.05},
       False:{True:.10,False:.95}}
p_a = {True:{True:.95,False:.01},
       False:{True:.05,False:.99}}
joint = {}
for f,s in product([True,False],repeat=2):
    joint[f,s] = p_f[f]*p_s[s][f]*p_a[True][s]
numerator = sum(joint[True,s] for s in [True,False])
denominator = sum(joint.values())
print(f"{numerator/denominator:.2%}")
```

HOW IT WORKS

Loop through every Fire and Smoke combination.

Multiply the three CPT factors.

Sum relevant rows and divide.

Expected: 23.46%.

EXACT INFERENCE

Why enumeration can become slow

WHAT IT MEANS

The number of assignments grows exponentially with the number of hidden binary variables.

With h hidden binary variables, there are 2^h combinations.

10 hidden variables → 1,024 cases.

20 → 1,048,576 cases.

MINI EXAMPLE

Enumeration is easy to understand, but large networks need smarter computation.

EXACT INFERENCE

Variable elimination: central idea

WHAT IT MEANS

Sum out hidden variables early, using reusable intermediate factors.

Instead of expanding all complete assignments:

1. Multiply factors containing a hidden variable.
2. Sum over that variable.
3. Keep the smaller resulting factor.

MINI EXAMPLE

Fire → Smoke → Alarm

Eliminate Smoke first to obtain a factor connecting Fire and Alarm.

Variable elimination: eliminate Smoke

Combine P(S|F) and P(A|S), then sum over S.

$$\varphi(F, A=ON) = \Sigma_s \, P(S|F) P(A=ON|S)$$

$\varphi(F=yes) = .90 \times .95 + .10 \times .01 = .856$

$\varphi(F=no) = .05 \times .95 + .95 \times .01 = .057$

We replaced two CPTs with one small factor φ(F).

Variable elimination: final calculation

Multiply the factor by the prior P(F), then normalize.

$$P(F|A) \propto P(F) \times \varphi(F,A)$$

F=yes: .02×.856=.01712

F=no: .98×.057=.05586

Normalize: .01712/(.01712+.05586)

Exact posterior = 23.46%, matching enumeration.

Python — variable elimination

RUN IN PYTHON OR GOOGLE COLAB

```
p_fire = {True:.02, False:.98}
p_smoke = {True:{True:.90,False:.10},
           False:{True:.05,False:.95}}
p_alarm = {True:.95, False:.01}
factor = {}
for fire in [True,False]:
    factor[fire] = sum(
        p_smoke[fire][smoke] * p_alarm[smoke]
        for smoke in [True,False]
    )
score = {f:p_fire[f]*factor[f] for f in factor}
posterior = score[True]/sum(score.values())
print(f"{posterior:.2%}")
```

HOW IT WORKS

1. Create the Smoke-eliminated factor.

2. Multiply it by the prior.

3. Normalize the two scores.

Expected: 23.46%.

Exact inference: what should students remember?

WHAT IT MEANS

Enumeration and variable elimination compute the same probability; they differ in efficiency.

Enumeration explicitly sums full assignments.

Variable elimination reuses smaller intermediate factors.

Exact inference may still be expensive on densely connected graphs.

MINI EXAMPLE

Same network, same CPTs, same answer: 23.46%.

WORKED EXAMPLE · TASK CONDITION

Task 4 — Compare exact and approximate inference

Use the same fire–smoke–alarm Bayesian Network for both methods.

GIVEN

Exact P(Fire | Alarm) ≈ 23.46%

Sampling produces a random estimate.

FIND

Why do the two methods differ in computation and sometimes numerical result?

SOLUTION PLAN

Exact: sum all hidden cases.

Approximate: simulate many cases and estimate frequencies.

WORKED EXAMPLE · PROBLEM STATEMENT

Task D — Estimate a probability by sampling

The alarm network is unchanged, but we will estimate P(Fire | Alarm) from simulated cases.

GIVEN

Fire prior = 2%

Smoke and alarm CPTs as before

Sample count = 10,000 or 100,000

FIND

How close can sampling get to the exact answer of 23.46%?

PLAN

Generate cases, retain alarm=ON, count how many contain a fire.

Sampling: intuition

WHAT IT MEANS

Sampling generates many plausible outcomes instead of summing every possible assignment.

Each simulated case follows the network's probabilities.

Repeated sampling approximates the joint distribution.

More samples often improve precision, but randomness remains.

MINI EXAMPLE

Simulate: Fire? → Smoke? → Alarm?

Repeat thousands of times.

Prior sampling: follow the arrows

WHAT IT MEANS

Generate variables in parent-before-child order.

1. Draw Fire from P(F).
2. Draw Smoke from P(S|F).
3. Draw Alarm from P(A|S).
4. Repeat.

MINI EXAMPLE

One possible sample:

Fire=no → Smoke=yes → Alarm=yes.

This sample is allowed by the CPTs.

Python — prior sampling

RUN IN PYTHON OR GOOGLE COLAB

```
import random
random.seed(42)
def sample_case():
    fire = random.random() < .02
    smoke = random.random() < (.90 if fire else .05)
    alarm = random.random() < (.95 if smoke else .01)
    return fire, smoke, alarm

samples = [sample_case() for _ in range(10000)]
print("Samples:", len(samples))
print("Alarms:", sum(a for f,s,a in samples))
```

HOW IT WORKS

random.random() draws a number from 0 to 1.

Compare it with a probability to sample a Boolean variable.

Each case respects the Bayesian Network.

EXACT INFERENCE

Rejection sampling: keep matching evidence

WHAT IT MEANS

To estimate P(Fire|Alarm), reject cases where Alarm is OFF.

1. Generate full cases using prior sampling.
2. Keep only Alarm=ON cases.
3. Estimate the fraction with Fire=YES.

MINI EXAMPLE

If 730 out of 10,000 cases have alarms, only those 730 cases contribute to the posterior estimate.

Python — rejection sampling

RUN IN PYTHON OR GOOGLE COLAB

```
import random
random.seed(42)
def draw():
    f = random.random() < .02
    s = random.random() < (.90 if f else .05)
    a = random.random() < (.95 if s else .01)
    return f,a

accepted = [f for f,a in (draw() for _ in range(100000))
            if a]
estimate = sum(accepted)/len(accepted)
print("Accepted:",len(accepted))
print(f"P(Fire|Alarm) ≈ {estimate:.2%}")
```

HOW IT WORKS

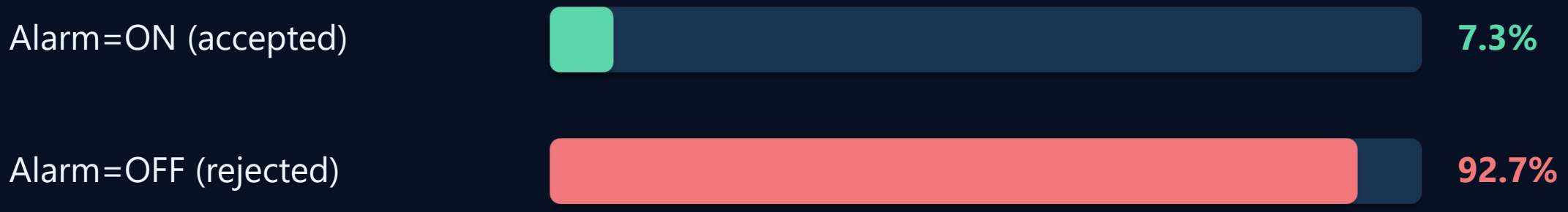
Discard cases that contradict Alarm=ON.

Count fire cases among accepted samples.

The estimate should approach 23.46% as samples increase.

SAMPLING

Why rejection sampling wastes samples



Only about 7.3% of prior samples match Alarm=ON; about 92.7% are discarded.

Likelihood weighting: avoid rejection

WHAT IT MEANS

Fix evidence variables and assign each sample a weight.

Do not randomly draw Alarm; force Alarm=ON.
Generate Fire and Smoke as usual.
Weight each case by P(Alarm=ON | Smoke).

MINI EXAMPLE

If Smoke=yes, weight=.95.
If Smoke=no, weight=.01.

No samples are discarded.

Python — likelihood weighting

RUN IN PYTHON OR GOOGLE COLAB

```
import random
random.seed(42)
yes_weight = total_weight = 0.0
for _ in range(100000):
    f = random.random() < .02
    s = random.random() < (.90 if f else .05)
    w = .95 if s else .01
    total_weight += w
    if f:
        yes_weight += w
print(f"{yes_weight/total_weight:.2%}")
```

HOW IT WORKS

1. Draw Fire and Smoke.

2. Fix Alarm=ON.

3. Weight by likelihood of observed evidence.

4. Normalize weighted counts.

EXACT INFERENCE

Gibbs sampling: the basic idea

WHAT IT MEANS

Gibbs sampling repeatedly resamples hidden variables conditioned on their neighbors and fixed evidence.

Initialize non-evidence variables.

Hold observed variables fixed.

Update one hidden variable at a time from its conditional distribution.

Discard warm-up samples and estimate frequencies.

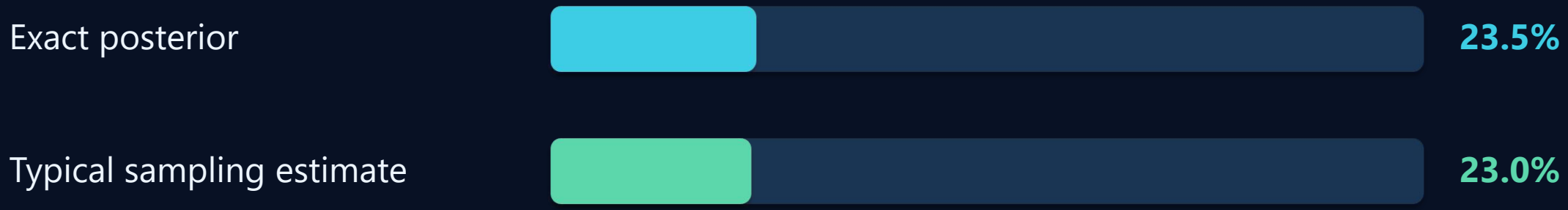
MINI EXAMPLE

With Alarm=ON, alternate updates to Fire and Smoke.

This is an MCMC method, not independent prior sampling.

SAMPLING

Exact vs sampling: compare the results



Illustrative sampling estimate shown; the actual estimate changes with the random seed and sample count.

Choosing the inference method

WHAT IT MEANS

Use the method that fits the network and computational constraints.

Enumeration: small networks, teaching, verification.

Variable elimination: structured exact inference.

Rejection sampling: simple but inefficient with rare evidence.

Likelihood weighting: useful with fixed evidence.

Gibbs sampling: iterative approximate inference.

MINI EXAMPLE

No single method is best for every Bayesian Network.

WORKED EXAMPLE · PROBLEM STATEMENT

Practice exercise — improve a detector

A machine has a 5% prior fault rate and 90% alarm sensitivity.

GIVEN

Scenario 1: false-alarm rate = 10%

Scenario 2: false-alarm rate = 2%

FIND

Compute P(Fault|Alarm) for both scenarios. Which detector is more trustworthy?

PLAN

Use Bayes' theorem and compare posteriors.

Practice solution — lower false alarms

Calculate the posterior for the 2% false-alarm scenario.

$$P(F|A) = .90\times.05 / (.90\times.05 + .02\times.95)$$

$= .045 / (.045 + .019) = .045 / .064 = .703125$

Posterior rises from 32.14% to 70.31%.

REVIEW

Summary — four questions to ask

WHAT IT MEANS

Every Bayesian inference problem becomes easier when you identify its structure.

1. What are the variables and arrows?
2. What evidence is observed?
3. What probability is the query?
4. Will we calculate exactly or estimate by sampling?

MINI EXAMPLE

Practice: write the task condition, list known CPT values, choose the method, and interpret the result.